

Project Design Phase
Proposed Solution Template

Date	15 February 2025
Team ID	LTVIP2025TMID32230
Project Name	SmartSDLC – AI-Enhanced Software Development Lifecycle
Maximum Marks	2 Marks

Proposed Solution Template:

Project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Traditional software development processes are time-consuming, manual, and error-prone. There's a lack of automation in requirement analysis and code generation.
2.	Idea / Solution description	SmartSDLC uses AI to automate key SDLC stages like requirement extraction from uploaded documents, automatic code generation, and test case generation.
3.	Novelty / Uniqueness	The solution integrates Hugging Face's LLMs with Streamlit and FastAPI to enable end-to-end automation of SDLC. It combines requirement analysis, code, and test generation in one interface.
4.	Social Impact / Customer Satisfaction	Developers, students, and project managers benefit from faster delivery, reduced manual work, and higher software quality, leading to greater satisfaction.
5.	Business Model (Revenue Model)	Freemium model: basic features free for individuals and students, premium features and API usage billed for enterprise teams and educational institutions.
6.	Scalability of the Solution	The platform can scale across domains—education, IT firms, and startups. It supports integration with cloud and can evolve with more LLMs and SDLC plugins.